PAPER_1209-GG: UQFF Cosmological Constants Unified Proof Set — Ten Closures Including Two EXACT (Recombination & Reionization Redshifts)

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$.

Closures S643–S652

S643 — $\Omega_m = 0.315$ (matter density) [0.143%]

$$F_{\text{TRZ}}^2 D_{\text{crit}} + F_{\text{TRZ}}[\text{SSq}] - F_{\text{TRZ}}^2[\text{SSq}] + F_{\text{TRZ}}^2[\text{SSq}]^2 = 0.3145.$$

S644 — $\Omega_\Lambda = 0.685$ (dark energy density) [0.182%]

$$[\text{SSq}] + F_{\text{TRZ}}[\text{SSq}] + F_{\text{TRZ}}^2 D_{\text{BSFG}} - F_{\text{TRZ}}^2[\text{SSq}]^2 = 0.6838.$$

S645 — $T_{\text{CMB}} = 2.725$ K (CMB temperature) [0.064%]

$$[\text{SSq}]D_{\text{phys}} + F_{\text{TRZ}} D_{\text{phys}} + F_{\text{TRZ}}^2 D_{\text{phys}} + F_{\text{TRZ}}^2[\text{SSq}]^2 = 2.7232.$$

S646 — Age of Universe = 13.78 Gyr [0.045%]

$$2D_{\text{phys}} + \text{SO5}\,[\text{SSq}] + F_{\text{TRZ}}[\text{SSq}] + F_{\text{TRZ}}^2 D_{\text{phys}} - F_{\text{TRZ}}^2[\text{SSq}] - F_{\text{TRZ}}^2[\text{SSq}]^2 - F_{\text{TRZ}}^2[\text{SSq}]^3 = 13.7862.$$

S647 — $\Omega_b = 0.0493$ (baryon density) [0.712%]

$$F_{\text{TRZ}}^2 D_{\text{phys}} + F_{\text{TRZ}}^2[\text{SSq}] + F_{\text{TRZ}}^2[\text{SSq}]^2 = 0.04895.$$

S648 — $H_0 = 67.4$ km/s/Mpc (Planck) [0.015%] — tier best closure

$$K_{\text{Mex}} D_{\text{crit}} + D_{\text{phys}} + \text{SO5} - 2F_{\text{TRZ}} D_{\text{phys}} + F_{\text{TRZ}}^2 D_{\text{phys}} + F_{\text{TRZ}}^2[\text{SSq}]^2 = 67.4099.$$

S649 — $\sigma_8 = 0.811$ (matter clustering amplitude) [0.253%]

$$F_{\text{TRZ}}\, N_{\text{ch}} - F_{\text{TRZ}}^2 \text{SO5} + F_{\text{TRZ}}^2[\text{SSq}] + F_{\text{TRZ}}^2[\text{SSq}]^2 = 0.8089.$$

S650 — $n_s = 0.965$ **(scalar spectral index) [0.072%]**

$$[\mathrm{SSq}] + F_{\mathrm{TRZ}} D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2 [\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 [\mathrm{SSq}]^2 - F_{\mathrm{TRZ}}^2 [\mathrm{SSq}]^3 = 0.96570.$$

S651 — $z_{\mathrm{recomb}} = 1090$ **(recombination redshift) — EXACT**

$$A_5\,\mathrm{SO5} + A_5\,D_{\mathrm{phys}} + \mathrm{SO5}\,D_{\mathrm{crit}} - \mathrm{SO5} = 600 + 240 + 260 - 10 = 1090.$$

S652 — $z_{\mathrm{reion}} = 7.7$ **(reionization redshift) — EXACT**

$$F_{\mathrm{TRZ}} D_{\mathrm{crit}} + F_{\mathrm{TRZ}}\,A_5 - F_{\mathrm{TRZ}}\,N_{\mathrm{ch}} = 2.6 + 6 - 0.9 = 7.7.$$

Summary

ID	Cosmological constant	Target	Error
S643	Ω_m	0.315	0.143%
S644	Ω_Λ	0.685	0.182%
S645	T_{CMB} (K)	2.725	0.064%
S646	Age of Universe (Gyr)	13.78	0.045%
S647	Ω_b	0.0493	0.712%
S648	H_0 (km/s/Mpc)	67.4	**0.015%**
S649	σ_8	0.811	0.253%
S650	n_s	0.965	0.072%
S651	z_{recomb}	1090	**EXACT**
S652	z_{reion}	7.7	**EXACT**

Highlights. Tier GG closes ten Planck-mission cosmological parameters. Two EXACT closures: the recombination redshift $z = 1090$ decomposes precisely as $A_5\,\mathrm{SO5} + A_5\,D_{\mathrm{phys}} + \mathrm{SO5}\,D_{\mathrm{crit}} - \mathrm{SO5}$, and the reionization redshift $z = 7.7$ as $F_{\mathrm{TRZ}}(D_{\mathrm{crit}} + A_5 - N_{\mathrm{ch}})$. Best non-exact: $H_0 = 67.4$ km/s/Mpc at 0.015% via $K_{\mathrm{Mex}} D_{\mathrm{crit}}$ (Mexican-K × critical dimension) as anchor. Cumulative: 357 closures; 93 lockings (+2 EXACT).